

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `802a19f5-2932-4b86-8002-6788fd7f7e5a`
- **Timestamp:** 2026-09-05T15:16:05.674092Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** In [2017] 13 S.C.R. 477, Bharat Petroleum Corporation Ltd. (BPCL) and Hindustan Petroleum A Corporation Ltd.
- **Answer:** I do not have enough information to answer this.
- **Health Score:** 0.08
- **Primary Issue:** RETRIEVAL_MISS
- **Claims Supported:** 0 / 1
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
---	---	---	---
CORPUS	UNKNOWN	0.5	Corpus relevance is inconclusive.
RETRIEVER	FAIL	0.85	No retrieved chunks provide support, and retrieval scores strongly indicate a lack of useful evidence.
GENERATOR	UNKNOWN	0.5	1 generated claims are unsupported, but retrieval quality is ambiguous.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 0/6 chunks used (0%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 1 (Supported: 0, Partial: 0, Unsupported: 1, Contradicted: 0)
- **Grounding Score:** 0.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.03
- **Latencies:** Retrieval (1267.0ms) | Generation (4577.0ms) | Verification (2402.0ms)

5. RAGAS Metrics

- **Faithfulness:** 0.0000
- **Answer Relevancy:** 0.3713
- **Context Precision:** 0.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[254b3584-7bb9-434a-9f1c-0b029408d76d]** (UNSUPPORTED): I do not have enough information to answer this.
- *Evidence (Chunk 5cfa6c8a-9305-43ab-81c3-730ca54163fc):* 1]

THE GAZETTE OF INDIA EXTRAORDINARY
35

7. Root Cause Analysis

- **Primary Cause:** RETRIEVAL_MISS
- **Confidence:** 0.88
- **Reasoning:**

- Starting root cause analysis for trace 802a19f5-2932-4b86-8002-6788fd7f7e5a.
- Skipping CORPUS: status is UNKNOWN (insufficient evidence).
- Failure at RETRIEVER: RETRIEVAL_MISS (confidence=0.85). Observation: No retrieved chunks provide support, and retrieval
- Skipping CHUNKING: no observable state in matrix.
- Skipping GENERATOR: status is UNKNOWN (insufficient evidence).
- GROUNDING passed (confidence=0.90).
- Primary root cause: RETRIEVAL_MISS at RETRIEVER (earliest upstream failure, confidence=0.85).
- Analysis complete. Primary cause: RETRIEVAL_MISS.

8. Corrective Action Plan

- ****[IMMEDIATE] Increase Retrieval Top-K****
 - Description: Increase the number of retrieved documents before providing context to the LLM.
 - Expected Improvement: Higher recall; missing context is caught within the expanded window. (Metric: Verification support)
- ****[SHORT_TERM] Implement Hybrid Search (Dense + Sparse)****
 - Description: Add BM25 lexical search and combine with semantic vector search using Reciprocal Rank Fusion.
 - Expected Improvement: Near-perfect recall for exact-match terminology. (Metric: Retrieval hit rate for exact entity queries)
- ****[EXPERIMENTAL] Deploy Cross-Encoder Reranker****
 - Description: Pass the Top-N results through a Cross-Encoder to re-score query-document relevance.
 - Expected Improvement: Drastically improved precision at Top-1 and Top-3. (Metric: Precision@3 increases by at least 20%)
- ****[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff****
 - Description: Retrieval efficiency is low (0%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- ****Primary Cause:**** RETRIEVAL_MISS
- ****Derived From:**** RootCauseAnalysis
- ****Pipeline Stage:**** RETRIEVAL_MISS
- ****Supporting Claims:**** 254b3584-7bb9-434a-9f1c-0b029408d76d
- ****Supporting Chunks:**** 5cfa6c8a-9305-43ab-81c3-730ca54163fc

10. Overall Assessment

- ****Major Strength:**** GROUNDING is healthy (confidence=90%): No claims are contradicted by retrieved evidence.
- ****Major Weakness:**** RETRIEVAL_MISS
- ****Next Priority:**** Increase Retrieval Top-K
- ****Recommendation:**** Pipeline requires architectural adjustments across multiple stages.

11. Framework Footer

- ****Generated By:**** X-RAG Diagnostic Framework
- ****Artifact Version:**** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/802a19f5-2932-4b86-8002-6788fd7f7e5a.json

✓ ClaimSet

artifacts/claim_sets/802a19f5-2932-4b86-8002-6788fd7f7e5a.json

✓ VerificationResults

artifacts/verification/802a19f5-2932-4b86-8002-6788fd7f7e5a.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/802a19f5-2932-4b86-8002-6788fd7f7e5a.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/802a19f5-2932-4b86-8002-6788fd7f7e5a.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/802a19f5-2932-4b86-8002-6788fd7f7e5a.json

✓ DiagnosticEvaluationReport

artifacts/reports/802a19f5-2932-4b86-8002-6788fd7f7e5a.json